

LUCIDPREP

Grade 8 Mathematics

U.S. Standards-Aligned Course Syllabus

MATH 8

Math 8 · Common Core

36 weeks • 4 modules • 70% mastery

Curriculum Architecture

This course is organized as 36 weeks across four 9-week modules. Every week follows LucidPrep's established secondary learning cycle: AI Instructor opening, objectives, vocabulary, five warm-ups, main lecture, two short tutorial videos, AI Instructor presentation, two homework sets, and a 40-question assessment.

Mastery standard: 70%+. Students below mastery review the instructional presentation and may retake the assessment, matching the existing LucidPrep secondary model.

U.S. Alignment Note

The U.S. has no single national curriculum, but Math 8 is the standard Grade 8 mathematics course in most districts. This syllabus uses the Common Core State Standards for Mathematics as the primary national reference and cross-checks against California Common Core and Texas TEKS, which cover the same content under different codes.

Prepared for LucidPrep • Curriculum Planning Edition • 2026

Common Instructional & Assessment Model

Weekly Learning Cycle

Step	Component	Description
1	Opening / Domain Overview	~2-minute AI Instructor introduction
2	Objectives + Vocabulary	Student-facing outcomes and core terminology
3	Warm-Up	5 questions with explanations
4	Main Lecture	Concept instruction, models and examples
5	Video Tutorials	2 short instructional videos
6	AI Instructor Presentation	Visual models, examples, strategies and applications
7	Homework	Two sets of 5 questions
8	Weekly Assessment	40 questions; 70% mastery
9	Review / Retake	Below mastery → review presentation → retake

Assessment Philosophy

Assessment is continuous and mastery-based. Each week pairs low-stakes practice (five warm-ups and two homework sets) with a 40-question weekly assessment. Students scoring below 70% review the instructional presentation and retake, so mastery is reached before progressing. Module assessments at Weeks 9, 18 and 27 and a cumulative final at Week 36 measure retention and synthesis.

MATH 8

Grade 8 mathematics aligned to
the Common Core State
Standards

This 36-week course is designed around **the Common Core Grade 8 domains: The Number System (8.NS), Expressions & Equations (8.EE), Functions (8.F), Geometry (8.G), and Statistics & Probability (8.SP)**. It is a full-year curriculum with four nine-week modules and a consistent LucidPrep mastery workflow.

Grade	Grade 8
Course	Math 8 · Common Core
Duration	36 weeks
Structure	4 modules × 9 weeks
Weekly assessment	40 questions
Mastery threshold	70%
Module assessments	Weeks 9, 18, 27
Final assessment	Week 36

Course Learning Outcomes

- Work with rational and irrational numbers and integer exponents.
- Analyze and solve linear equations and systems of equations.
- Define, evaluate, compare and model with functions.
- Apply transformations, congruence, similarity and the Pythagorean theorem.
- Investigate bivariate data and patterns of association.

MATH 8**Module 1 — Number System & Exponents**

Week	Topics	Students will be able to	Standards
1 — Rational & Irrational Numbers	Definitions, decimal expansions, estimating irrationals	Distinguish rational and irrational numbers	8.NS.1/2
2 — Approximating & Comparing Irrationals	Locating on a number line; comparing	Compare and order real numbers	8.NS.2
3 — Integer Exponents	Product, quotient and power rules; zero/negative exponents	Apply properties of integer exponents	8.EE.1
4 — Square & Cube Roots	Perfect squares/cubes, solving $x^2=p$ and $x^3=p$	Evaluate and use roots	8.EE.2
5 — Scientific Notation	Representing, comparing and interpreting magnitudes	Express numbers in scientific notation	8.EE.3
6 — Operations in Scientific Notation	Computing with very large/small numbers	Operate with scientific notation	8.EE.4
7 — Applications of Exponents & Roots	Real-world contexts and measurement	Apply exponents and roots to problems	8.EE.1-4
8 — Number-System Reasoning	Mixed reasoning across the number system	Reason across the number system	8.NS/8.EE
9 — Module 1 Assessment	Cumulative review of the number system	Demonstrate mastery of Module 1	8.NS/8.EE

Module delivery notes. Build toward Algebra 1 by centering linear relationships and functions. Use graphs, tables and equations together, and connect geometry to algebraic reasoning.

MATH 8**Module 2 — Linear Equations & Systems**

Week	Topics	Students will be able to	Standards
1 — Solving Linear Equations	One variable, distributive, variables both sides	Solve multi-step linear equations	8.EE.7
2 — Equations with Special Solutions	One, none or infinitely many solutions	Determine the number of solutions	8.EE.7a
3 — Proportional Relationships & Slope	Unit rate as slope; comparing relationships	Interpret slope from proportional relationships	8.EE.5
4 — Slope & y-Intercept	Deriving $y = mx + b$; similar triangles and slope	Relate slope and intercept to graphs	8.EE.6
5 — Graphing Linear Equations	Graphing from equations and tables	Graph linear equations	8.EE.6; 8.F.3
6 — Systems by Graphing	Solutions as points of intersection	Solve systems by graphing	8.EE.8a/b
7 — Systems — Algebraic Methods	Substitution and elimination	Solve systems algebraically	8.EE.8b
8 — Systems Applications	Real-world problems with two equations	Model and solve with systems	8.EE.8c
9 — Module 2 Assessment	Cumulative review of equations and systems	Demonstrate mastery of Module 2	8.EE

Module delivery notes. Build toward Algebra 1 by centering linear relationships and functions. Use graphs, tables and equations together, and connect geometry to algebraic reasoning.

Module 3 — Functions

Week	Topics	Students will be able to	Standards
1 — Introduction to Functions	Definition, input/output, function vs not	Identify functions	8.F.1
2 — Function Representations	Tables, graphs, equations, verbal descriptions	Translate among representations	8.F.1
3 — Comparing Functions	Comparing properties across representations	Compare two functions	8.F.2
4 — Linear Functions	$y = mx + b$, rate of change, initial value	Interpret linear functions	8.F.3/4
5 — Modeling with Linear Functions	Constructing from a description or two points	Model relationships with linear functions	8.F.4
6 — Nonlinear Functions	Recognizing nonlinear (e.g., $y = x^2$)	Distinguish linear and nonlinear functions	8.F.3
7 — Analyzing Graphs Qualitatively	Increasing/decreasing; describing relationships	Describe and sketch graphs from context	8.F.5
8 — Applications of Functions	Real-world modeling and interpretation	Apply functions to problems	8.F.4/5
9 — Module 3 Assessment	Cumulative review of functions	Demonstrate mastery of Module 3	8.F

Module delivery notes. Build toward Algebra 1 by centering linear relationships and functions. Use graphs, tables and equations together, and connect geometry to algebraic reasoning.

Module 4 — Geometry & Data

Week	Topics	Students will be able to	Standards
1 — Transformations	Translations, reflections, rotations; coordinates	Describe transformations	8.G.1/3
2 — Congruence	Congruent figures via transformations	Establish congruence with transformations	8.G.2
3 — Similarity & Dilations	Dilations, similar figures, angle-angle	Describe similarity with dilations	8.G.4
4 — Angle Relationships	Parallel lines/transversals, triangle angle sum	Use angle relationships	8.G.5
5 — Pythagorean Theorem	Understanding and applying the theorem	Apply the Pythagorean theorem	8.G.6/7
6 — Pythagorean Distance & 3D	Distance between points; 3D applications	Find distance and apply in three dimensions	8.G.8
7 — Volume of Curved Solids	Cones, cylinders, spheres	Compute volumes of curved solids	8.G.9
8 — Bivariate Data	Scatter plots, association, line of best fit, two-way tables	Investigate patterns of association	8.SP.1-4
9 — Final Assessment	Cumulative final across all modules	Demonstrate mastery of the full course	8.NS-8.SP

Module delivery notes. Build toward Algebra 1 by centering linear relationships and functions. Use graphs, tables and equations together, and connect geometry to algebraic reasoning.

Assessment & Alignment

Assessment Blueprint

Warm-ups: weekly • Homework: weekly • Weekly assessment: 40 questions • Module assessments: Weeks 9 / 18 / 27 • Final: Week 36 • Below 70%: instructional review and retake.

Standards Alignment

Primary framework	CCSS: 8.NS, 8.EE, 8.F, 8.G, 8.SP
Mathematical Practices	Make sense of problems and persevere; reason abstractly and quantitatively; construct viable arguments and critique reasoning; model with mathematics; use tools strategically; attend to precision; use structure and regularity.
Recurring Concepts	Multiple representations; equivalence; proportional reasoning; structure; modeling; real-world application.
State mapping	Common Core adopted by the large majority of states; Texas TEKS and other state frameworks cover the same grade-level content under different codes.

Content Production Guidance

Author each week as 5 warm-up items, a concept lecture with worked examples and multiple representations, 2 curated videos, an AI presentation with models and strategies, 2 homework sets (5 items each), and a 40-question bank spanning fluency, application and reasoning. Tag every item to its Common Core code.

Curriculum Basis & Source Notes

This Grade 8 mathematics syllabus was generated from the U.S. Common Core State Standards for Mathematics. Content and pacing reflect the standard Math 8 · Common Core course used across most U.S. districts.

Primary reference	Common Core State Standards for Mathematics — Grade 8 domains and the Standards for Mathematical Practice.
State cross-check	California Common Core State Standards for Mathematics.
State cross-check	Texas Essential Knowledge and Skills (TEKS) for Mathematics.
LucidPrep alignment	Mapped to LucidPrep's 36-week, 4-module secondary structure with 70% mastery.

Recommended next production step. Commission item banks per week (40 assessment items plus warm-ups and homework), each tagged to its Common Core code, then load into the LucidPrep authoring pipeline.